

Software Requirements and Sustainability

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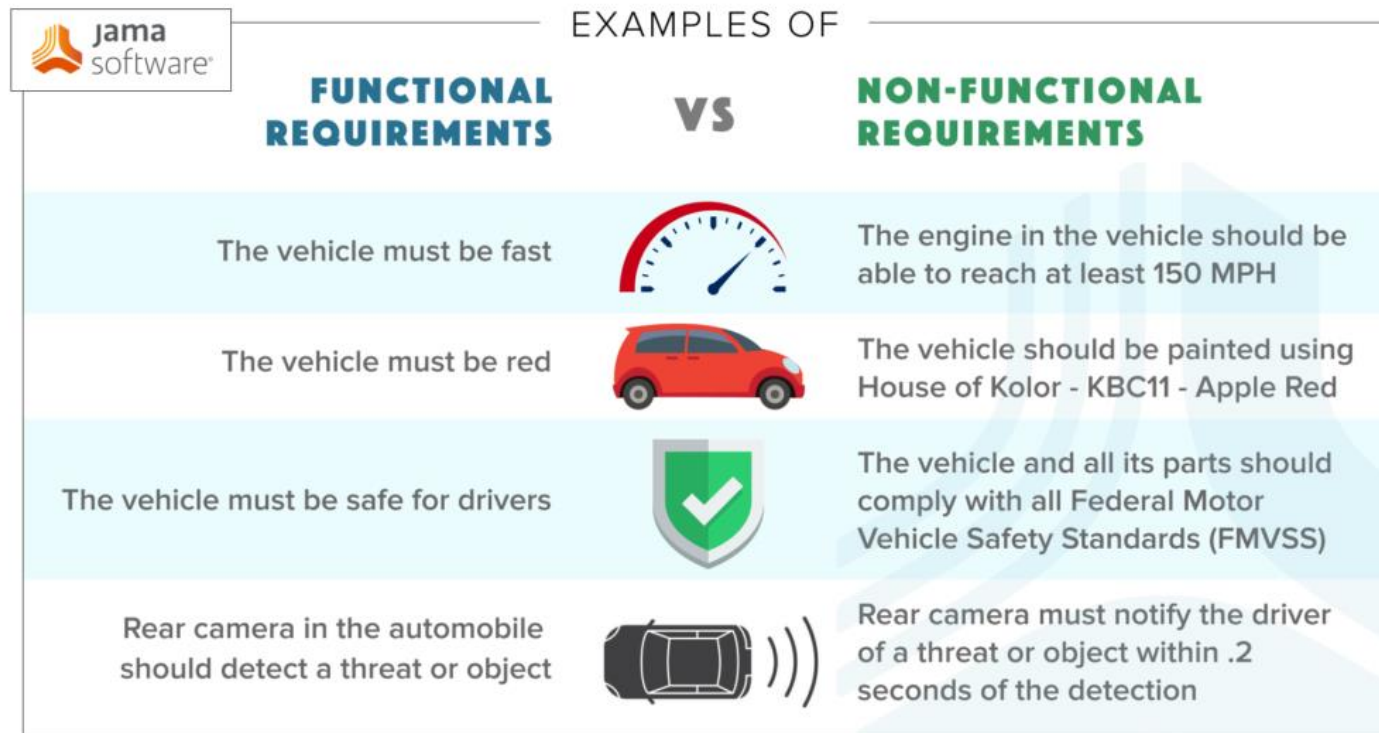
Software Requirements and Sustainability

Functional Requirements:

- Describe what the system should do and are related to user needs.

Non-Functional Requirements:

- Define how the system performs its functions.



Sustainability in Software Systems

Sustainability Dimensions in Software:

- 1. Social:** Impact on society and communities.
Encourage inclusivity and accessibility.
- 2. Environmental:** Reducing negative environmental impacts. Optimize resource use and reduce waste.
- 3. Individual:** Enhance user well-being, satisfaction, user experience and privacy protection.
- 4. Technical:** Long-term adaptability and evolution.
Ensure scalability, maintainability, and resilience.
- 5. Economic:** Balancing financial benefits and costs.
Reduce costs and promote financial efficiency.

Foundation (1): 5 dimensions of sustainability



[GIBSE'13] Birgit Penzenstadler, Henning Femmer
A Generic Model for Sustainability with Process- and Product-specific Instances.
1st Intl. Workshop on Green In Software Engineering, Green By Software Engineering (at AOSD'13)

Gather Requirements with Sustainability Considerations

Identify Stakeholders:

- Include users, developers, sustainability experts, and business analysts.

Use Elicitation Techniques:

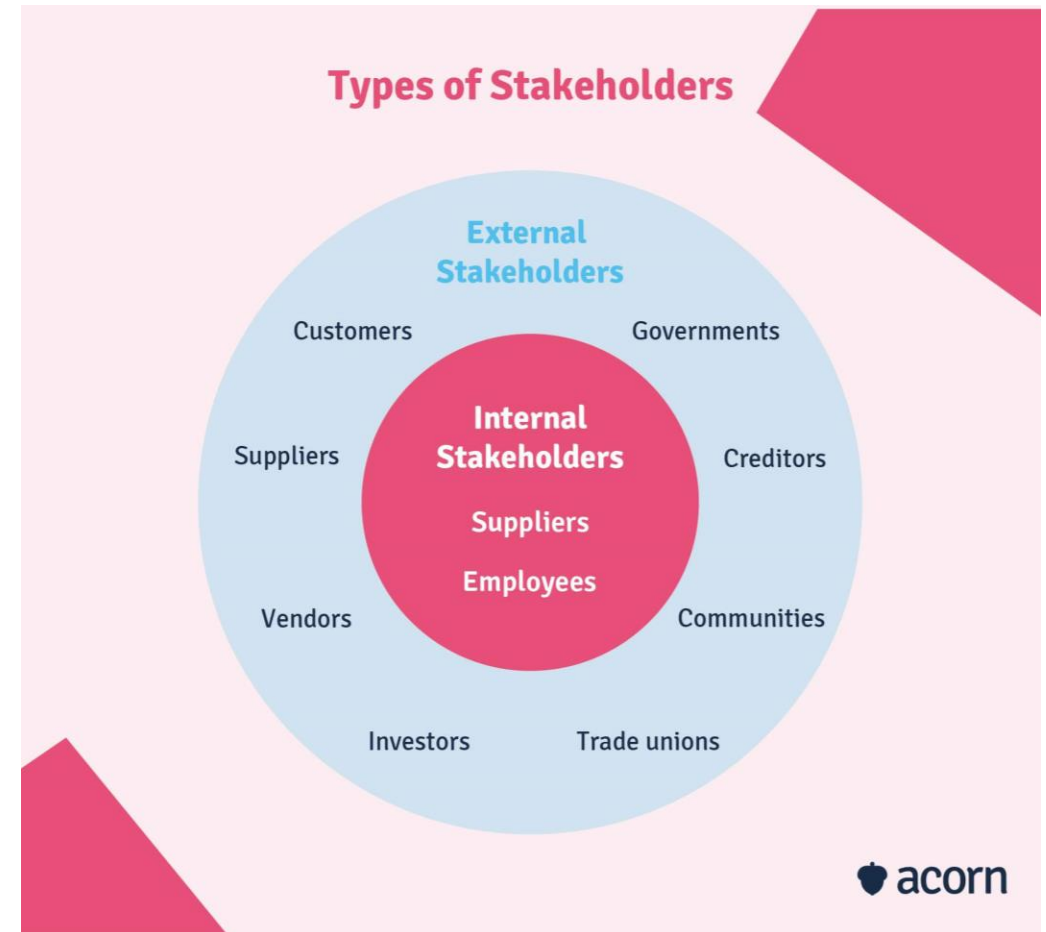
- Surveys, interviews, workshops, observations, and brainstorming sessions that include sustainability questions.

Focus on Sustainability Goals:

- Determine how each requirement can impact social, environmental, technical, and economic sustainability.

Document Requirements:

- Use clear, concise language that captures sustainability goals and ensures traceability and alignment with sustainability.



Writing Requirements with Sustainability Focus

Template for Functional Requirements:

- "The system shall [action] [sustainability impact]."
- **Example:** "The system shall adjust video resolution based on bandwidth to reduce energy usage."

Template for Non-Functional Requirements:

- "The system shall achieve [performance metric] to ensure [sustainability goal]."
- **Example:** "The system shall ensure a minimum of 70% renewable energy use in data centers to reduce environmental impact."

A university plans to develop a **Sustainability Awareness Application** aimed at educating software engineering students about the environmental impacts of software systems and IT services. The app will feature interactive modules, quizzes, and personalized recommendations for eco-friendly software development practices.

Your team is tasked with gathering detailed requirements to ensure the app meets the following goals:

- Engages students with interactive learning.
- Provides accurate sustainability content.
- Offers tailored recommendations based on user behavior.
- Integrates seamlessly into students' existing digital platforms (e.g., LMS or mobile).

Functional Requirements with Sustainability

- The system shall deliver interactive learning modules to increase students' engagement with sustainability concepts.
- The system shall track user interactions and learning progress to provide personalized recommendations that promote eco-friendly software development habits.
- The system shall provide quizzes that assess students' understanding of sustainable software engineering to encourage knowledge retention and behavioral change.
- The system shall integrate with existing Learning Management Systems (LMS) to minimize duplication of resources and reduce digital waste.

Non-Functional Requirements with Sustainability

- The system shall achieve at least 90% uptime to ensure continuous access to sustainability learning resources with minimal disruption.
- The system shall ensure that 80% of its data processing occurs on energy-efficient cloud infrastructure to reduce carbon footprint.
- The system shall maintain a response time below 2 seconds to minimize unnecessary computation and improve user experience efficiency.
- The system shall ensure at least 95% accuracy of sustainability content to guarantee reliable and evidence-based learning.
- The system shall achieve integration compatibility with at least three major LMS platforms (e.g., Moodle, Canvas, Blackboard) to encourage reuse and reduce development redundancy.

Netflix Case Example

Functional Requirements with Sustainability

Example 1: Energy-Efficient Streaming

- **Requirement:** The system shall dynamically adjust video resolution based on the user's network bandwidth to minimize data usage.
 - **Sustainability Impact:** Reduces energy consumption by decreasing data transmission and processing load.

Example 2: Localized Content for Diverse Audiences

- **Requirement:** The system shall provide content recommendations based on user preferences and region to promote cultural diversity.
 - **Sustainability Impact:** Supports social inclusion by promoting culturally relevant content to diverse audiences.

Example 3: Smart Downloading Feature

- **Requirement:** The system shall allow users to download content in off-peak hours to minimize network congestion.
 - **Sustainability Impact:** Reduces energy usage during peak times by balancing network load.

Functional Requirements with Sustainability

4. Accessibility for Disabled Users

- **Requirement:** The system shall provide audio descriptions and closed captions for all content.
 - **Sustainability Impact:** Enhances social sustainability by improving accessibility for disabled users.

5. Regional Language Support

- **Requirement:** The system shall provide content in at least three local languages for each supported region.
 - **Sustainability Impact:** Supports social inclusivity by providing access to diverse populations.

6. Renewable Energy Usage Tracking

- **Requirement:** The system shall provide analytics to track energy consumption from renewable sources for each streaming session.
 - **Sustainability Impact:** Encourages transparency and supports green energy adoption.

7. Eco-Friendly Content Recommendations

- **Requirement:** The system shall recommend low-resolution content in areas with limited bandwidth to minimize data consumption.
 - **Sustainability Impact:** Reduces data transmission, saving bandwidth and energy.

Non-Functional Requirements with Sustainability

Example 1: Green Data Centers

- **Requirement:** The system shall run on data centers powered by at least 70% renewable energy sources.
 - **Sustainability Impact:** Minimizes the carbon footprint of Netflix's operations.

Example 2: Privacy-Conscious Personalization

- **Requirement:** The system shall provide personalized content recommendations without storing unnecessary sensitive personal data.
 - **Sustainability Impact:** Protects individual privacy while improving user trust and satisfaction, contributing to long-term user retention.

3. High-Performance Scalability

- **Requirement:** The system shall be scalable to handle a 30% increase in users without significant energy cost increases.
 - **Sustainability Impact:** Ensures long-term efficiency as the user base grows.

Non-Functional Requirements with Sustainability

4. Low Carbon Footprint Infrastructure

- **Requirement:** The system shall ensure a 25% reduction in carbon emissions from cloud infrastructure by 2025.
 - **Sustainability Impact:** Promotes environmental sustainability by reducing the environmental footprint.

5. Support for Older Devices

- **Requirement:** The system shall support video streaming on devices manufactured within the last 10 years.
 - **Sustainability Impact:** Extends the life cycle of older devices, reducing electronic waste.

6. Energy-Efficient User Interface

- **Requirement:** The system shall provide a dark mode option to reduce energy consumption on OLED displays.
 - **Sustainability Impact:** Reduces energy usage by lowering the power needed for display lighting.

Best Practices for Gathering Requirements with Sustainability Consideration

- **Engage Diverse Stakeholders:** Include sustainability experts, users, and developers in the process.
- **Quantify Sustainability Goals:** Set measurable objectives, like reducing energy consumption by 20% or reduce carbon emissions by 25%.
- **Monitor and Adapt:** Continuously assess and update sustainability goals based on changing needs, technology advances and and market trends.
- **Prioritize Sustainability Early:** Address sustainability from the start of the project to ensure alignment with development goals.

Questions?